

1. Inventory Cost Structure – The Basics

Every company that keeps stock (inventory) has costs. For EOQ, we focus on the two costs that *change* with how much and how often you order:

- **Ordering Costs** (also called setup or procurement costs): These are the fixed costs you pay **every single time** you place an order with a supplier. Examples: paperwork, emails/calls, delivery arrangement, inspection fees. → It doesn't matter if you order 10 units or 1,000 units — the cost per *order* stays the same (e.g. \$10 per order). More orders = higher total ordering cost.
- **Holding Costs** (also called carrying or storage costs): These are the costs of *keeping* items in your warehouse. Examples: warehouse rent/space, insurance, electricity, money you can't use elsewhere (opportunity cost), risk of damage/obsolescence/theft. → This is charged **per unit per year**. The more units you hold and the longer you hold them, the higher the cost.

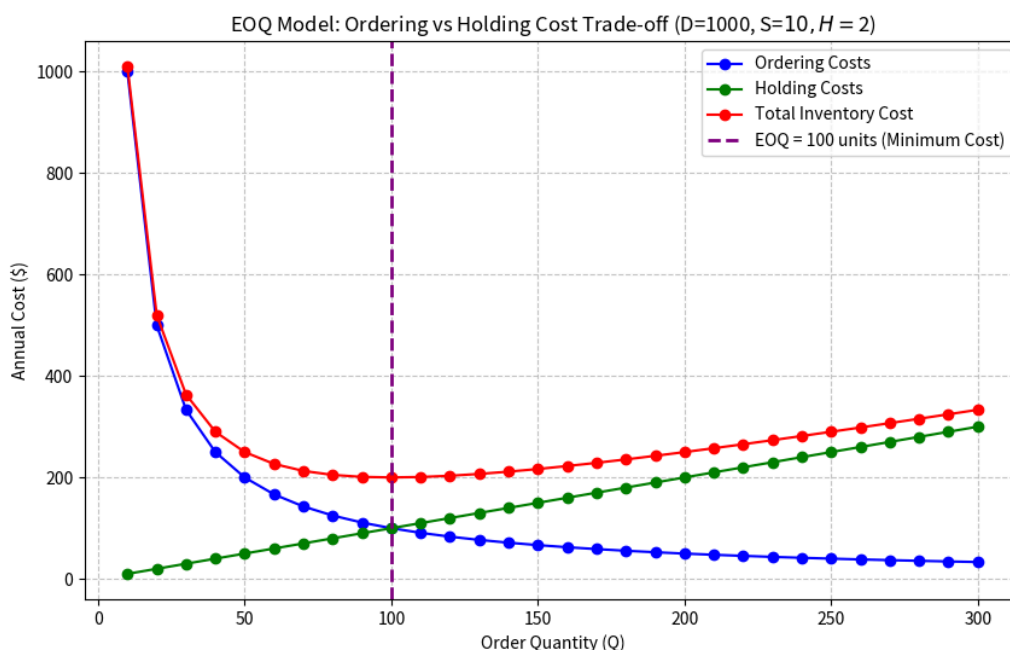
(Note: The actual purchase price of the goods is usually ignored in basic EOQ because it stays the same no matter the order size. We'll add it later if there are discounts.)

Total relevant inventory cost (per year) = Ordering cost + Holding cost

2. Learn: Ordering vs Holding Cost Trade-off

This is the heart of EOQ — a classic seesaw.

- **Small orders + frequent ordering** (e.g. order 10 units every few days): → You place *lots* of orders → **high ordering costs**. → But you keep very little stock sitting around → **low holding costs**.



- **Large orders + rare ordering** (e.g. order 1,000 units once a year): → You place *few* orders → **low ordering costs**. → But you have tons of stock sitting in the warehouse on average → **high holding costs**.

If you plot these on a graph, ordering cost falls as order size grows, holding cost rises, and the *total* cost makes a U-shape. The lowest point of the U is the "sweet spot" — the cheapest way to manage inventory.

Here's a clear visual of that trade-off (using example numbers: 1,000 units demand per year, \$10 ordering cost, \$2 holding cost per unit).

3. What is EOQ and the Formula?

Economic Order Quantity (EOQ) is the *perfect* order quantity that gives you the lowest total inventory cost.

The simple formula is:

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$$EOQ = \sqrt{\frac{2 \times D \times S}{H}}$$

- D = Annual demand (units you expect to sell/use in a year)
- S = Ordering cost per order (\$)
- H = Holding cost per unit per year (\$)

Quick example (we'll use this all day):

D = 1,000 units/year

S = \$10 per order

H = \$2 per unit per year

$$EOQ = \sqrt{\frac{2 \times 1000 \times 10}{2}} = \sqrt{10000} = 100 \text{ units}$$

- You order 100 units each time → 10 orders per year (1,000 ÷ 100).
- Average inventory = 50 units (half the order sits there on average).
- Total cost = (10 orders × \$10) + (50 units × \$2) = \$100 + \$100 = **\$200/year**.

